

- Based on the following information arrange four metals A, B, C and D in order of decreasing ability to act as reducing agents :
 (I) Only A, B and C react with 1 M HCl to give $H_2(g)$
 (II) When C is added to solutions of the other metal ions, metallic B and D are formed
 (III) Metal C does not reduce A^{n+} .
 (A) $C > A > B > D$ (B) $A > D > C > B$ (C) $B > D > A > C$ (D) $A > C > B > D$
- A solution containing H^+ and D^+ ions is in equilibrium with a mixture of H_2 and D_2 gases at $25^\circ C$. If the partial pressures of both the gases be 1.0 atm, find the ratio of $[D^+]/[H^+]$:
 (Given : $E_{D^+/D_2}^0 = -0.003 V$)
 (A) 1.23 (B) 1.12 (C) 0.11 (D) 1.0
- The standard electrode potential for the following reaction is +1.33 V. What is the potential at pH = 2.0 ?
 $Cr_2O_7^{2-}(aq, 1 M) + 14 H^+(aq) + 6e^- \longrightarrow 2Cr^{3+}(aq, 1 M) + 7H_2O(l)$
 (A) +1.820 V (B) +1.990 V (C) +1.608 V (D) +1.0542 V
- The electrolysis of acetate solution produces ethane according to reaction :
 $2CH_3COO^- \longrightarrow C_2H_6(g) + 2CO_2(g) + 2e^-$
 The current efficiency of the process is 80%. What volume of gases would be produced at $27^\circ C$ and 740 torr, if the current of 0.5 amp is passed through the solution for 96.45 min ?
 (A) 6.0 L (B) 0.60 L (C) 1.365 L (D) 0.91 L
- 100 mL of 0.05 M $CuSO_4(aq)$ solution was electrolyzed using inert electrodes by passing current till the pH of the resulting solution was 2. The solution after electrolysis was neutralized and then treated with excess KI and formed I_2 titrated with 0.04 M $Na_2S_2O_3$. Calculate the required volume (in mL) of $Na_2S_2O_3$:
 (A) 112.5 mL (B) 100 mL (C) 125 mL (D) None of these
- If a 100 mL solution of 0.1 M HBr is titrated using a very concentrated solution of NaOH, then the conductivity (specific conductance) of this solution at the equivalence point will be (assume volume change is negligible due to addition of NaOH). Report your answer after multiplying it with 10 in $S m^{-1}$. Assume a very little variation in molar conductance of NaBr between given concentration and at infinite dilution.
 [Given $\lambda_{(Na^+)}^0 = 8 \times 10^{-3} Sm^2 mol^{-1}$, $\lambda_{(Br^-)}^0 = 4 \times 10^{-3} Sm^2 mol^{-1}$]
 (A) 6 (B) 12 (C) 15 (D) 24

7. For electrolyte MCl_2 in water at 25°C the following data is given:
- | | | |
|---|--------|-----|
| conc mole/lit | : 0.16 | 1 |
| $\lambda_m (\Omega^{-1}\text{cm}^2\text{mol}^{-1})$ | : 282 | 240 |
- calculate λ_m^∞ at 25°C in $\Omega^{-1}\text{cm}^2\text{mol}^{-1}$
- (A) 300 (B) 320 (C) 310 (D) 290
8. The standard reduction potential of a silver chloride electrode is 0.2 V and that of a silver electrode is 0.79 V.
- The maximum amount of AgCl that can dissolve in 10^6 L of a 0.1 M AgNO_3 solution is
- (A) 0.5 mmol (B) 1.0 mmol (C) 2.0 mmol (D) 2.5 mmol
9. Standard electrode potential of two half-reactions are given below
- $\text{Fe}^{2+} \rightleftharpoons \text{Fe}; E^\circ = -0.44\text{ V}$
- $\text{Fe}^{3+} \rightleftharpoons \text{Fe}^{2+}; E^\circ = +0.77\text{ V}$
- If Fe^{2+} , Fe^{3+} and Fe are kept together,
- (A) The concentration of Fe^{3+} increases. (B) The concentration of Fe^{3+} decreases.
- (C) The mass of Fe increases. (D) The concentration of Fe^{2+} decreases.

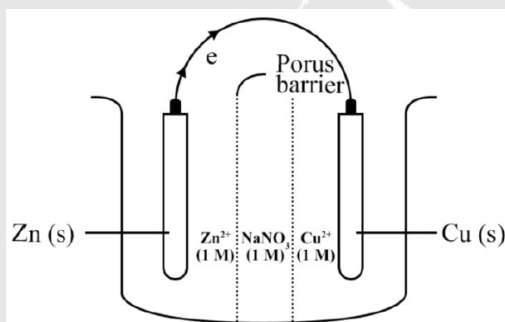
MULTIPLE CHOICE QUESTIONS

10. Given : $\text{Pt(s)} | \text{H}_2(\text{g}) | 0.1\text{ M NH}_4\text{OH(aq)} || 0.1\text{ M CH}_3\text{COOH(aq)} | \text{H}_2(\text{g}) | \text{Pt(s)}$
- 1 bar 1 bar
- $\text{pK}_b(\text{NH}_4\text{OH}) = 5; \text{pK}_a(\text{CH}_3\text{COOH}) = 5; \frac{2.303RT}{F} = 0.06$
- Volume of 0.1 M NH_4OH in anode half cell = 100 mL.
- Volume of 0.1 M CH_3COOH in cathode half cell = 100 mL.
- Which is/are correct statement ?
- (A) The emf of given cell is 0.48 V.
- (B) The emf of given cell is 0.36 V when 50 mL, 0.1 M NaOH added to cathode compartment
- (C) The emf of given cell is 0.36 V when 50 mL, 0.1M HCl added to anode compartment
- (D) The emf of given cell is 0.192 V when 100 mL 0.1M NaOH added to anode compartment
11. Consider the cell $\text{Pt} | \text{H}_2(p_1\text{ atm}) | \text{H}^+(x_1\text{ M}) || \text{H}^+(x_2\text{ M}) | \text{H}_2(p_2\text{ atm}) | \text{Pt}$. The cell reaction will be spontaneous if
- (A) $p_1 = p_2$ and $x_1 > x_2$ (B) $p_1 = p_2$ and $x_1 < x_2$ (C) $x_1 = x_2$ and $p_1 > p_2$ (D) $x_1 = x_2$ and $p_1 < p_2$
12. Which of the following statement is correct?
- If $E^\circ_{\text{Cu}^{2+}|\text{Cu}} = 0.34\text{ V}$ and $E^\circ_{\text{Sn}^{2+}|\text{Sn}} = 0.136\text{ V}$, $E^\circ_{\text{H}^+|\text{H}_2} = 0.0\text{ V}$
- (A) Cu^{2+} ions can be reduced by $\text{H}_2(\text{g})$
- (B) Cu can be oxidized by H^+
- (C) Sn^{2+} ions can be reduced by H_2
- (D) Sn can be oxidized by Cu^{2+}

13. Which is/are correct statement?
- (A) No corrosion takes place in vacuum
- (B) Corrosion is protected by electroplating
- (C) During rusting $\text{Fe}_2\text{O}_3 \cdot x\text{H}_2\text{O}$ formed
- (D) In presence of electrolyte, corrosion takes place with greater rate

COMPREHENSION # 3 (FOR Q. 14 TO Q.16)

A Galvanic cell consist of three compartments as shown in figure. The first compartment contains ZnSO_4 (1 M) and III compartment contain CuSO_4 (1M). The mid compartment contains NaNO_3 (1 M). Each. compartments contain 1 L solution: $E^\circ_{\text{Zn}^{2+}/\text{Zn}} = -0.76$; $E^\circ_{\text{Cu}^{2+}/\text{Cu}} = +0.34$



14. The concentration of Zn^{2+} in first compartment after passage of 0.1 F charge will be?
- (A) 1 M (B) 1.05 M (C) 1.025 M (D) 0.5 M
15. The concentration of NO_3^- in mid compartment after passage of 0.1 F of charge will be?
- (A) 0.95 M (B) 0.90 M (C) 0.975 M (D) 1.05 M
16. The concentration of SO_4^{2-} ion in III compartment after passage of 0.1F of charge will be?
- (A) 1.05 M (B) 1.025 M (C) 0.95 M (D) 0.975 M

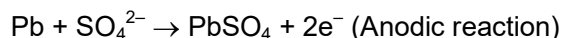
NUMERIC ANSWER TYPE

17. Calculate the emf of the cell in mV (at least first two digits must match with correct answer)
- $\text{Ag (s), AgIO}_3\text{(s)} \mid \text{Ag}^+\text{(xM), HIO}_3\text{(1M)} \parallel \text{Zn}^{+2}\text{(1M)} \mid \text{Zn (s)}$

If $K_{\text{sp}} = 3 \times 10^{-8}$ for AgIO_3 and $K_a = \frac{1}{6}$ for HIO_3 and E°_{cell} for $2\text{Ag} + \text{Zn}^{+2} \rightarrow 2\text{Ag}^+ + \text{Zn}$ is -1.56

K. ($\log 3 = 0.48$) (Take $\frac{RT}{F} = 0.059$) (giving your answer in magnitude only)

18. During the discharge of a lead storage battery, the density of sulphuric acid fell from 1.294 to 1.139 g ml^{-1} . H_2SO_4 of density 1.294 gm L^{-1} is 39% and that of density 1.139 gm L^{-1} is 20% by weight. The battery holds 3.5 L of acid and the volume practically remains constant during discharge. Calculate the number of ampere hours for which the battery must have been used. The discharging reactions are:



19. The conductivity of 0.1 M solution of a weak monoprotic acid is $2.19 \times 10^{-4} \text{ S cm}^{-1}$. The limiting molar conductivity of it is $400 \text{ S cm}^2 \text{ mol}^{-1}$. If the dissociation constant of the acid is $y \times 10^{-6}$, what is the value of y ?
20. Molar conductivity of aqueous solution of HA is $200 \text{ S cm}^2 \text{ mol}^{-1}$, pH of this solution is 4. Calculate the value of pK_a (HA) at 25°C .

Given : $\Lambda_M^\infty(\text{NaA}) = 100 \text{ S cm}^2 \text{ mol}^{-1}$; $\Lambda_M^\infty(\text{HCl}) = 425 \text{ S cm}^2 \text{ mol}^{-1}$; $\Lambda_M^\infty(\text{NaCl}) = 125 \text{ S cm}^2 \text{ mol}^{-1}$

21. Match the Following:

Column-I	Column-II
(A) $\text{Au} \left \text{AuCl}_4^- (1\text{M}) \right \left \text{Li}^+ (1\text{M}) \right \text{Li}$	(p) Spontaneous
(B) $\text{Zn} \left \text{Zn}^{2+} (1\text{M}) \right \left \text{Al}^{3+} (1\text{M}) \right \text{Al}$	(q) Non-spontaneous
(C) $\text{Cu} \left \text{Cu}^{2+} (C_1) \right \left \text{Cu}^{2+} (C_2) \right \text{Cu}$ $C_1 > C_2$	(r) $E^\circ_{\text{cell}} > 0$
(D) $\text{Ni} \left \text{Ni}^{2+} (1\text{M}) \right \left \text{Cu}^{2+} (1\text{M}) \right \text{Cu}$	(s) $E^\circ_{\text{cell}} < 0$
	(t) galvanic cell
(A) (A - s; B - q, s; C - p; D - p, r)	(B) (A - q, s; B - q, s; C - q; D - p, r, t)
(C) (A - q; B - q; C - q; D - t)	(D) (A - p, q; B - r, s; C - q; D - t)

22. Column-I and Column-II contains four entries each. Entries of Column-I are to be matched with some entries of Column-II. One or more than one entries of Column-I may have the matching with the same entries of Column-II

Column-I	Column-II
(A) Dilute solution of HCl	(P) O_2 evolved at anode
(B) Dilute solution of NaCl	(Q) H_2 evolved at cathode
(C) Concentrated solution of NaCl	(R) Cl_2 evolved at anode
(D) Fairly concentrated solution of AgNO_3	(S) Ag deposition at cathode
(A) $A \rightarrow P, Q; B \rightarrow R, Q; C \rightarrow Q, R; D \rightarrow P, S$	
(B) $A \rightarrow P, Q; B \rightarrow P, Q; C \rightarrow Q, R; D \rightarrow P, Q$	
(C) $A \rightarrow P, Q; B \rightarrow R, Q; C \rightarrow Q, R; D \rightarrow P, S$	
(D) $A \rightarrow P, Q; B \rightarrow P, Q; C \rightarrow Q, R; D \rightarrow P, S$	

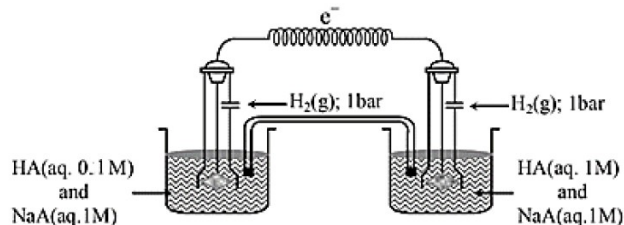
23. In the Hall process, aluminum is produced by the electrolysis of molten Al_2O_3 . How many seconds would it take to produce enough aluminum by the Hall process to make a case of 24 cans of aluminum soft drink, if each can use 5.0 g of Al, a current of 9650 amp is employed, and the current efficiency of the cell is 90.0%.

(A) 203.2 (B) 148.14 (C) 333 (D) 6.17

24. Copper reduces NO_3^- into NO and NO_2 depending upon concentration of HNO_3 in solution. Assuming $[\text{Cu}^{2+}] = 0.1 \text{ M}$, and $P_{\text{NO}} = P_{\text{NO}_2} = 10^{-3} \text{ bar}$. At which concentration of HNO_3 , thermodynamic tendency for reduction of NO_3^- into NO and NO_2 by copper is same?

[Given : $E^\circ_{\text{Cu}^{2+}|\text{Cu}} = +0.3\text{volt}$, $E^\circ_{\text{NO}_3^-|\text{NO}} = +0.96\text{volt}$, $E^\circ_{\text{NO}_3^-|\text{NO}_2} = +0.79\text{volt}$]

- (A) $10^{1.23} \text{ M}$ (B) $10^{0.56} \text{ M}$ (C) $10^{0.66} \text{ M}$ (D) $10^{0.12} \text{ M}$
25. A cell diagram shown below contains one litre of buffer solution of HA ($\text{pK}_a = 4$) and NaA in both compartments. What is the cell EMF?



- (A) 0.03 V (B) 0.06 V (C) -0.06 V (D) None of these

SUBJECTIVE ANSWER TYPE

26. Determine potential (in mV) of the cell : $\text{Pt} | \text{Fe}^{2+}, \text{Fe}^{3+} || \text{Cr}_2\text{O}_7^{2-}, \text{Cr}^{3+}, \text{H}^+ | \text{Pt}$ in which $[\text{Fe}^{2+}] = 0.75 \text{ M}$, $[\text{Fe}^{3+}] = 0.75 \text{ M}$, $[\text{Cr}_2\text{O}_7^{2-}] = 2 \text{ M}$, $[\text{Cr}^{3+}] = 4 \text{ M}$ and $[\text{H}^+] = 1 \text{ M}$.

Given : $\text{Fe}^{3+} + \text{e}^- \longrightarrow \text{Fe}^{2+}$; $E^\circ = 0.77 \text{ V}$

$14 \text{ H}^+ + 6\text{e}^- + \text{Cr}_2\text{O}_7^{2-} \longrightarrow 2\text{Cr}^{3+} + 7\text{H}_2\text{O}$; $E^\circ = 1.35 \text{ V}$

$2.303 \text{ RT/F} = 0.06$, $\log 2 = 0.3$, $\log 3 = 0.48$

27. A sample of water from a large swimming pool has a resistance of 9200Ω at 25°C when placed in a certain conductance cell. When filled with 0.02 M KCl solution, the cell has a resistance of 85Ω at 25°C . 500 gm of NaCl were dissolved in the pool, which was thoroughly stirred. A sample of this solution gave a resistance of 7600Ω . Calculate the volume of water in the pool.

Given: Molar conductance of NaCl at that concentration is $126.5 \Omega^{-1} \text{ cm}^2 \text{ mol}^{-1}$ and molar conductivity of KCl at 0.02 M is $138 \Omega^{-1} \text{ cm}^2 \text{ mol}^{-1}$.